

PROJECT PROPOSAL : Data Science Capstone Project

FRAUD DETECTION FOR FINANCIAL RISK MANAGEMENT

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1. OVERALL DATASET STRUCTURE

- We are using the **Online Payments Fraud Detection Dataset** for this project.
- The dataset contains simulated financial transactions and is designed for studying fraudulent and legitimate transaction behavior.
- The dataset contains approximately **6.36 million transaction records and 11 attributes**.
- The data represents transactions occurring over a simulated period of approximately 30 days.
- Each transaction contains information related to:
    - Transaction time
    - Transaction type
    - Transaction amount
    - Originating customer
    - Originating customer balance
    - Destination customer
    - Destination customer balance
    - Fraud status
    - Fraud detection flag
- The primary objective is to use the transaction information to determine whether a transaction is **fraudulent or legitimate**.

2. DATASET ATTRIBUTES

Dataset – Online Payments Fraud Detection

The dataset contains the following major attributes:

1. Step

- Represents the time of the transaction.
- Each step represents approximately one hour.
- The dataset contains up to 744 hourly steps representing the simulated transaction period.

2. Type

- Represents the type of transaction.
- The major transaction types include:
    - CASH-IN
    - CASH-OUT
    - DEBIT
    - PAYMENT
    - TRANSFER

3. Amount

- Represents the amount of money involved in the transaction.
- This attribute will be analyzed to determine whether transaction size is associated with fraudulent activity.

4. nameOrig

- Represents the customer who initiated the transaction.
- This is an identifier field and will be evaluated during preprocessing to determine whether it provides useful predictive information.

5. oldbalanceOrg

- Represents the originating customer's balance before the transaction.

6. newbalanceOrig

- Represents the originating customer's balance after the transaction.

7. nameDest

- Represents the customer receiving the transaction.
- Similar to `nameOrig`, this is an identifier field and will be evaluated during preprocessing.

8. oldbalanceDest

- Represents the destination customer's balance before the transaction.

9. newbalanceDest

- Represents the destination customer's balance after the transaction.

10. isFraud

- Represents the target variable for the project.
- A value of **0 represents a legitimate transaction**.
- A value of **1 represents a fraudulent transaction**.

11. isFlaggedFraud

- Represents whether a transaction was flagged by the original rule-based fraud detection mechanism.
- This attribute will be examined during the exploratory and feature analysis stages.

The dataset's documented feature structure includes these 11 attributes, with `isFraud` serving as the fraud classification target.

3. INITIAL DATA ANALYSIS

The first stage of the project involves understanding and preparing the transaction dataset.

Students will:

- Load the dataset using Python.
- Examine the overall structure of the dataset.
- Identify the number of rows and attributes.
- Check the data types of each attribute.
- Check for missing or null values.
- Identify duplicate transactions.
- Examine the distribution of transaction types.
- Analyze transaction amounts.
- Examine sender and receiver balance patterns.
- Analyze the distribution of fraudulent and legitimate transactions.
- Identify potential outliers.
- Examine relationships between transaction attributes and fraud.

A major challenge in this dataset is **class imbalance**. The dataset contains 8,213 fraudulent transactions out of 6,362,620 total transactions, meaning fraudulent transactions represent only about **0.13%** of all transactions.

Therefore, accuracy alone will not be considered an appropriate measure of model performance.

4. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis will be performed to understand patterns associated with fraudulent transactions.

The analysis will include:

- Distribution of transaction types.
- Fraudulent transactions by transaction type.
- Transaction amount distributions.
- Comparison of transaction amounts between fraud and legitimate transactions.
- Sender balance analysis.

- Receiver balance analysis.
- Fraud transactions over time.
- Correlation analysis among numerical variables.
- Identification of potentially important features.
- Visualization of fraud versus non-fraud transactions.

The results of the EDA will be used to identify patterns and guide the feature engineering and machine learning stages of the project.

5. DATA PREPROCESSING AND FEATURE ENGINEERING

The dataset will be prepared before developing the machine learning models.

The preprocessing stage will include:

- Handling missing values if present.
- Checking and removing duplicate records where appropriate.
- Converting categorical variables into suitable numerical representations.
- Evaluating identifier fields such as `nameOrig` and `nameDest`.
- Scaling numerical variables where required by the selected models.
- Identifying and handling potential outliers.
- Separating predictor variables from the target variable.

Feature engineering will be performed to create additional variables that may better represent transaction behavior.

Potential engineered features may include:

- Difference between old and new sender balances.
- Difference between old and new destination balances.
- Transaction amount relative to sender balance.
- Transaction amount relative to destination balance.
- Transaction time-based features derived from `step`.
- Transaction-type indicators.
- Balance consistency features.

The usefulness of these engineered features will be evaluated before they are included in the final models.

6. MACHINE LEARNING MODEL DEVELOPMENT

The final analysis will focus on developing and comparing classical machine learning models for fraud classification.

A. Random Forest

- Train a Random Forest classification model.
- Analyze feature importance.
- Tune important model parameters where appropriate.
- Evaluate its ability to identify fraudulent transactions.
- Examine its performance under the highly imbalanced class distribution.

B. Gradient Boosted Trees / XGBoost

- Develop an XGBoost classification model.
- Tune important hyperparameters.
- Compare its performance with Random Forest.
- Examine important features contributing to fraud predictions.

Random Forest and XGBoost are selected because tree-based models can capture nonlinear relationships between transaction characteristics and fraud behavior. The original project plan also identifies these two models as the primary classification methods.

7. HANDLING CLASS IMBALANCE

Class imbalance is one of the most important challenges in this project.

Since fraudulent transactions represent only a very small percentage of the complete dataset, a model could achieve high overall accuracy while failing to identify fraudulent transactions.

To address this issue, the project will investigate appropriate techniques such as:

- Class weighting.
- Random undersampling.
- Appropriate sampling strategies.
- Threshold adjustment.
- Other imbalance-handling techniques where appropriate.

The effectiveness of these techniques will be compared using fraud-focused evaluation metrics rather than relying only on accuracy.

8. MODEL EVALUATION

The developed models will be evaluated using metrics appropriate for highly imbalanced classification problems.

The primary evaluation metrics will include:

Precision

- Measures the proportion of transactions predicted as fraud that are actually fraudulent.

Recall

- Measures the proportion of actual fraudulent transactions correctly identified by the model.

F1-Score

- Provides a balance between precision and recall.

AUPRC

- Area Under the Precision-Recall Curve.
- This will be an important metric because of the highly imbalanced nature of the dataset.

Confusion Matrix

- Used to examine:
    - True Positives
    - True Negatives
    - False Positives
    - False Negatives

The models will be compared using these metrics to determine which approach provides the most useful fraud detection performance.

9. MODEL INTERPRETABILITY

Model interpretability will be an important part of the project because fraud detection systems should provide understandable reasons behind their predictions.

The project will investigate:

- Feature importance from Random Forest.
- Feature importance from XGBoost.
- SHAP-based explanations where appropriate.
- Identification of the transaction characteristics most strongly associated with fraud.
- Individual transaction-level explanations where feasible.

This analysis will help explain why a machine learning model identifies a particular transaction as potentially fraudulent.

SHAP has been identified in the project plan as a potential method for interpreting fraud detection models.

10. FINAL GOALS AND EVALUATION

The project will be considered successful if it demonstrates:

Technical Accuracy

- Development of effective machine learning models for fraud classification.
- Meaningful identification of fraudulent transactions.
- Comparison between Random Forest and XGBoost.
- Evaluation using precision, recall, F1-score, AUPRC, and confusion matrices.

Data Science Process

- Proper data cleaning and preprocessing.
- Meaningful exploratory data analysis.
- Appropriate handling of class imbalance.
- Effective feature engineering.
- Proper model training and evaluation.

Explainability

- Identification of important features affecting fraud predictions.
- Clear explanation of model behavior.
- Use of feature importance and SHAP or similar techniques where appropriate.

Communication

- Comprehensive documentation of the data science workflow.
- Clear visualizations of important findings.
- Presentation of model results and comparisons.
- Final written report and project presentation.

11. EXPECTED PROJECT OUTCOMES

At the completion of the project, we expect to:

- Develop a complete fraud detection machine learning pipeline.
- Identify important transaction characteristics associated with fraud.
- Compare Random Forest and XGBoost models.
- Determine suitable methods for handling the highly imbalanced fraud dataset.
- Identify the model that provides the most useful balance between fraud detection and false alarms.
- Explain the most important factors contributing to model predictions.
- Provide visualizations and documented findings that communicate the results clearly.

12. REQUIRED TECHNOLOGIES AND LIBRARIES

Programming Language

1. Python

Data Processing

2. Pandas
3. NumPy

Machine Learning

4. Scikit-learn
5. XGBoost

Data Visualization

6. Matplotlib
7. Seaborn

Model Explainability

8. SHAP
9. LIME, if required

Development Environment

10. Jupyter Notebook / Google Colab

The original project plan identifies Python, pandas, scikit-learn, NumPy, XGBoost, Matplotlib/Seaborn, and SHAP/LIME as the primary technical requirements.